Project Update

Numerical Simulation of Manufacturing Process

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1. Progress
2. Currently working on
3. Future work

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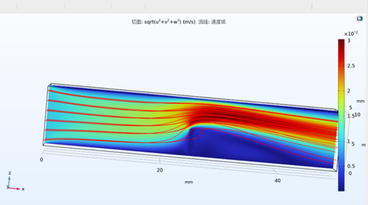
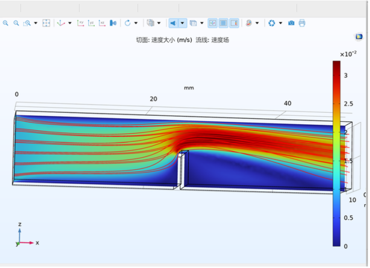
3. Future work

First, we validated the proposed method by comparing it with the body-fitted mesh model. The results show that both methods yield identical flow patterns, with only minor discrepancies in some details.

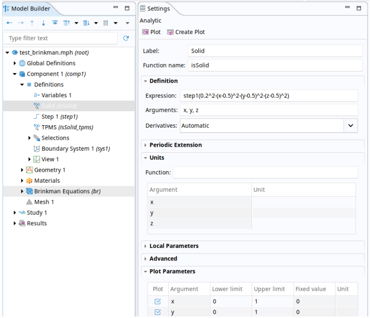


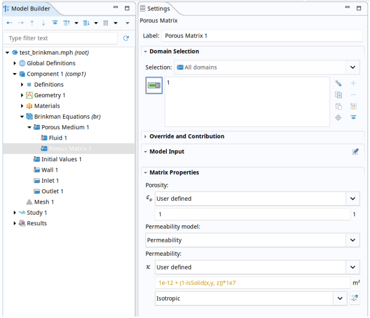


Sphere example

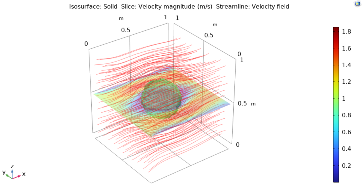
Before considering a TPMS structure, we have to be familiar with Comsol and Brinkman penilization. For Axel, this means starting with a more simple example.

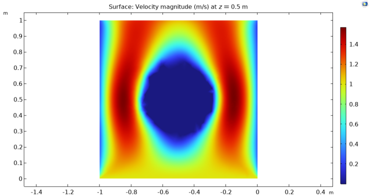
- Define an analytic function for the obstacle
- Define the porous medium properties for the region





Sphere example results



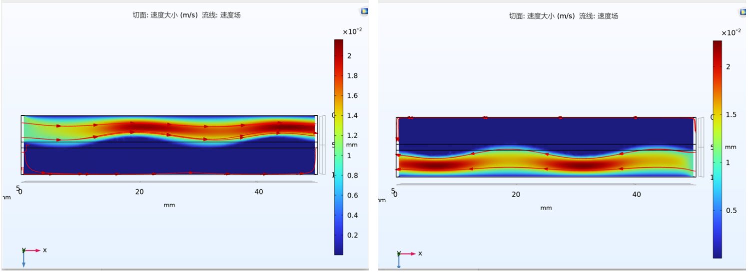


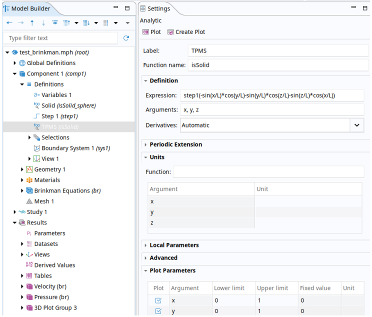
Conclusion: Brinkman frameworks works for very simple geometries

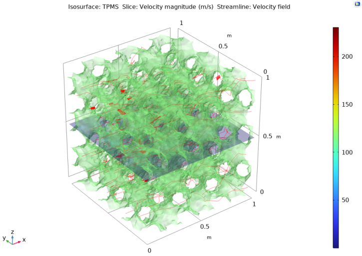
TPMS analysis

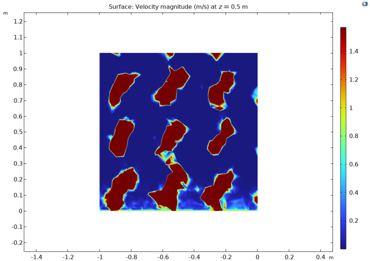
The gyroid TMPS structure can be implemented in the same way as the sphere, i.e. by an analytic expression.

Subsequently, we implemented the dual-channel model, which requires two sets of governing equations. The general approach is as follows: when calculating Fluid 1, the domain of Fluid 2 and all wall boundaries are set as high-resistance regions. Presented below are the results of this approach verified by a simple example.




Model Builder
Type filter text
test_brinkman.mph (root)
Global Definitions
Component 1 (comp1)
Definitions
Variables 1
Solid (isSolid_sphere)
Step 1 (step1)
TPMS (isSolid)
Selections
Boundary System 1 (sys1)
View 1
Geometry 1
Materials
Brinkman Equations (br)
Mesh 1
Study 1
Results
Parameters
Datasets
Views
Derived Values
Tables
Velocity (br)
Pressure (br)
3D Plot Group 3
Settings
Analytic
Plot
Create Plot
Label: TPMS
Function name: isSolid
Definition
Expression: step1(-sin(x/L)*cos(y/L)-sin(y/L)*cos(z/L)-sin(z/L)*cos(x/L))
Arguments: x, y, z
Derivatives: Automatic
Periodic Extension
Units
Function:
Argument
Unit
x
y
z
Local Parameters
Advanced
Plot Parameters
Plot
Argument
Lower limit
Upper limit
Fixed value
Unit
x 0 1 0
y 0 1 0


Isosurface: TPMS Slice: Velocity magnitude (m/s) Streamline: Velocity field
200
150
100
50


Surface: Velocity magnitude (m/s) at z = 0.5 m
1.4
1.2
1
0.8
0.6
0.4
0.2

- Incorporate heat transfer into the model
- Successfully implement a two-channel model